



Project Requirements for Revenue and Occupancy Analysis – 365 Hotels & Resorts

Overview

This project covers a full revenue and occupancy analysis for **365 Hotels & Resorts**, a hospitality group operating 25 properties across Dubai, Abu Dhabi, Sharjah and Ras Al Khaimah, split between Luxury and Business categories. Revenue reaches us through seven booking channels, from our own direct site to third party aggregators.

Occupancy has been soft for two consecutive quarters. The revenue team believes the weakness sits in a specific mix of city, room class and booking channel, but so far nobody has put numbers to that belief. The purpose of this analysis is to replace opinion with evidence and identify exactly where occupancy and revenue are leaking.

The work is to be delivered as a **single Jupyter notebook in Python**, supported by a short written summary of findings. The notebook must run top to bottom without errors and the summary must be readable by someone who does not write code, because it goes to the General Manager.

Your Role

You are joining as a Data Analyst reporting directly to the Revenue Manager. You have been given three months of booking data covering May to July 2022, plus one week of August data that has just landed from the properties. Your job is to tell us where occupancy and revenue are actually being lost, backed by numbers that will hold up in front of the GM.

1. Data Understanding and Setup

- You have been provided with six files. Review all of them before writing a single line of analysis code.

File	Grain	Approx. Rows
dim_hotels.csv	One row per property	25
dim_rooms.csv	One row per room class	4
dim_date.csv	One row per day, May to July 2022	92
fact_aggregated_bookings.csv	Property x date x room class	9,200
fact_bookings.csv	One row per individual booking	134,590
new_data_august.csv	Early August extract, pre-joined	7

- fact_bookings** is the detailed transaction table. **fact_aggregated_bookings** is the daily capacity table. They answer different questions and a good analysis uses both.



- Work out which columns join to which before you begin. Sketch the schema on paper if that helps. Getting the joins wrong quietly corrupts everything downstream.
- Set up your notebook with a clear structure: imports, data loading, cleaning, metrics, analysis, visuals, summary. Keep the sections separated with markdown headers.

2. Exploratory Analysis

- For the aggregated bookings table:
 - Confirm how many distinct properties appear and whether that matches `dim_hotels`.
 - Report total bookings per property and identify which properties carry the volume.
 - Identify property and date combinations where `successful_bookings` exceeds capacity, and explain in one sentence what such a row could mean operationally.
 - Identify the properties with the highest total capacity, and state whether high capacity actually translates into high bookings.
- For the detailed bookings table:
 - Compare the date range of `booking_date` against `check_in_date` and quantify how far ahead guests are booking.
 - Break down `booking_status` and report what share of bookings never convert into a stay.

3. Data Cleaning Requirements

This data comes out of four different property management systems and the inconsistencies show. Every cleaning decision must be documented with the reasoning behind it.

- **Missing values.** Two columns carry nulls: **capacity** in the aggregated table and **ratings_given** in the bookings table.
 - Fill capacity using a sensible approach and justify the choice you made.
 - For ratings_given, do not impute before investigating. Roughly 58% of the column is null. Examine which bookings are missing a rating first, then write one paragraph on whether imputing the mean is defensible here. There is a correct answer and it is not "fill with the mean".
- **Impossible values.** The **no_guests** column contains values that cannot physically exist. Locate them and decide whether to drop or correct them.
- **Outliers.** The **revenue_generated** column has a small number of values orders of magnitude above everything else. Find them, decide whether they are data entry errors or genuine bulk bookings, and state how you reached that decision.
- **Over-capacity rows.** Remove rows where `successful_bookings` exceeds capacity. Report how many rows were dropped and what percentage of the table that represents. Always report what you throw away.



4. Core Metric Development

Build the following as reusable columns or functions rather than one-off calculations. You will need them repeatedly across the analysis.

- **Occupancy %** = $\text{successful_bookings} / \text{capacity}$. This is the single most important number in hospitality and everything else sits downstream of it.
- **ADR (Average Daily Rate)** = $\text{total realized revenue} / \text{number of completed stays}$. What an occupied room actually earns.
- **RevPAR (Revenue Per Available Room)** = $\text{ADR} \times \text{Occupancy \%}$. This is the metric that catches the trap of chasing occupancy by cutting rates. A hotel can be full and still be failing.
- **Realization %** = $\text{revenue_realized} / \text{revenue_generated}$. Group this by `booking_status`. The pattern you find is a business rule; state it in plain language.
- **Length of Stay** = `checkout_date` minus `check_in_date`. You will need proper date types for this, not strings.

5. Required Analysis

Each of the following needs a number and a sentence explaining what that number means for the business.

- **Occupancy**
 - Average occupancy by city, identifying which city is underperforming.
 - Average occupancy by room class, and whether the pattern holds across all four cities or one city breaks it.
 - Occupancy on weekdays versus weekends. The gap is large; state what it implies about who these hotels are actually serving.
 - Occupancy by week across the 12 weeks, with a clear statement on whether the trend is up, down or flat.
- **Revenue**
 - Total realized revenue by hotel category, Luxury versus Business.
 - RevPAR by property, ranked, naming the top three and bottom three.
 - A comparison of the revenue ranking against the occupancy ranking, flagging any property that is high occupancy but low RevPAR and explaining what is happening there.
- **Booking Channels**
 - Revenue and booking volume by `booking_platform`.
 - Cancellation rate by platform. Look carefully before writing your conclusion. If the rates are broadly the same, say so. A finding of "no difference" is a real finding and reporting it honestly is worth more than manufacturing a pattern that is not there.



- A recommendation on which channel deserves further investment, supported by the two points above.
- **Guest Satisfaction**
 - Average rating by city, using only bookings that actually carry a rating.
 - Whether rating correlates with occupancy at property level. Compute it, then state whether the relationship is strong enough to act on.

6. Visualization Requirements

Six charts are required. Every chart needs a title, labelled axes, and a one line takeaway written underneath it in a markdown cell. A chart with no stated takeaway is decoration.

- Bar chart: occupancy % by city, split by room class.
- Line chart: weekly occupancy trend across the 12 weeks.
- Grouped bar chart: weekday versus weekend occupancy for each city.
- Horizontal bar chart: RevPAR by property, sorted.
- Bar chart: revenue by booking platform. Do not use a pie chart. With seven categories a pie makes the comparison harder, and comparing angles is worse than comparing bar lengths.
- Box plot: revenue_realized by hotel category, showing the spread and any remaining outliers.

7. Appending the August Data

The file **new_data_august.csv** arrives pre-joined and in a different shape from the other tables. Treat it as a test of whether your pipeline can absorb new data without breaking.

- Compare its columns against the structure you have been working with and document what matches and what does not.
- Reshape it so it can append cleanly to your aggregated bookings analysis.
- Verify the property attributes it carries agree with dim_hotels. Never trust attribute columns that arrive duplicated inside a fact file; always check them against the dimension.
- Append it and confirm the row count increased by the expected amount.

Instructions for the Analyst

- Marks and review comments go to reasoning, not line count. A short notebook with clear decisions and honest caveats beats a long one that computes everything and concludes nothing.
- Two things will specifically count against you: dropping rows without reporting how many, and stating a correlation as if it were a cause.
- Organise the notebook so a reader can follow the logic without you sitting next to them. Group related work together and use markdown cells to explain what you are about to do and why.



- Where a decision could reasonably have gone the other way, say so and explain the trade-off. We would rather see judgement than false certainty.

Deliverables

- A single Jupyter notebook that runs top to bottom without errors, covering all sections above.
- A written summary at the end of the notebook containing atleast three findings you would put in front of the GM.
- One specific, actionable recommendation with the number that supports it.
- A full record of every cleaning decision you made and the reasoning behind each one.
- A clear statement of what you could not answer with this data and what additional data you would need to answer it. Knowing the limits of your data is part of the job.

This analysis will feed directly into our rate strategy and channel investment decisions for the coming quarter, so please make it thorough, clearly reasoned and easy to interpret for both commercial and operational stakeholders. If anything in the data does not make sense, come and ask rather than guessing.

Kind regards,

Muhammad Abbas

Revenue Manager, Commercial Analytics

365 Hotels & Resorts